

STATISTICS 652

LECTURE 4-26

REVIEW OF THEOREM 3.6 AND PROOF.

STATEMENT: log-likelihood for an exponential family in natural form
IS STRICTLY CONCAVE IN $\eta \in \bar{\eta}$ ($\bar{\eta}$ CONVEX)

SKETCH OF PROOF; (CONTINUOUS CASE)

• $l(\eta, \eta) = \eta' T(\eta) + S(\eta) + d_0(\eta)$ IS STRICTLY CONCAVE IFF
 $d_0(\eta)$ IS STRICTLY CONCAVE (SINCE $\eta' T(\eta) + S(\eta)$ IS LINEAR)

• EXPRESS $d_0(\eta) = -\log \int_A \exp\{\eta' T(\eta) + S(\eta)\} d\eta$ AND

$$d_0((1-\lambda)\eta_0 + \lambda\eta_1) = -\log E_{\eta_1}[\exp\{(1-\lambda)(\eta_0 - \eta_1)' T(\eta)\}] + d_0(\eta_1)$$

• BY JENSEN'S INEQUALITY OR HOLDERS INEQUALITY

$$E[Y^{1-\lambda}] < E[Y]^{1-\lambda} \text{ for } Y \geq 0, P(Y=y) \neq 1 \text{ all } y.$$

• SO, $d_0((1-\lambda)\eta_0 + \lambda\eta_1) > -\log E_{\eta_1}[\exp\{(\eta_0 - \eta_1)' T(\eta)\}]^{1-\lambda} + d_0(\eta_1)$

$$= (1-\lambda)d_0(\eta_0) + \lambda d_0(\eta_1). \quad \square$$

FOR AN EXPONENTIAL FAMILY WITH NATURAL PARAMETER SPACE $\bar{\eta}$, PROVE

LEMMA 4.1: THE MGF EXISTS FOR ANY INTERIOR POINT $\eta \in \bar{\eta}$, WITH

$$M_X(\underline{s}) = E[e^{\underline{s}' X}] = \exp\{d_0(\underline{s} + \eta) - d_0(\eta)\} \quad \underline{s} \in N(\eta) \subset \mathbb{R}^k$$

LEMMA 4.2: THE DENSITY/MASS OF $T(\eta)$, THE NATURAL MINIMAL SUFFICIENT

STATISTIC IS GIVEN BY

$$f_T(\underline{t}) = \exp\{\eta' \underline{t} + d_0(\eta) + S(\underline{t})\} I_{A_0}(\underline{t}),$$

PROOF: HW (PROVE 4,2 FOR DISCRETE CASE ONLY)

RECALL: LIKELIHOOD EQUATION HAS AT MOST ONE-ROOT

$$\ell(\eta; \mathcal{X}) = \log L(\eta; \mathcal{X}) = \sum_{j=1}^k \eta_j T_j(\mathcal{X}) + d_0(\eta) + S(\mathcal{X})$$

$$\frac{\partial \ell}{\partial \eta_j} = T_j(\mathcal{X}) + \frac{\partial d_0(\eta)}{\partial \eta_j}$$

ROOT OF LIKELIHOOD EQUATION:

$$\nabla d_0(\eta) = -T(\mathcal{X})$$

IF A SOLUTION EXISTS IN $\mathcal{Y}$ THEN IT IS THE UNIQUE MLE.

Now:

$$\frac{\partial d_0(\eta)}{\partial \eta_j} = \frac{\partial}{\partial \eta_j} \left[-\log \int_A \exp\{\eta' T(\mathcal{X}) + S(\mathcal{X})\} d\mathcal{X} \right]$$

$$= - \frac{1}{\int_A \exp\{\eta' T(\mathcal{X}) + S(\mathcal{X})\} d\mathcal{X}} \int_A T_j(\mathcal{X}) \exp\{\eta' T(\mathcal{X}) + S(\mathcal{X})\} d\mathcal{X}$$

$$= - \int_A T_j(\mathcal{X}) \exp\{\eta' T(\mathcal{X}) + d_0(\eta) + S(\mathcal{X})\} d\mathcal{X} = -E_{\eta}[T_j(\mathcal{X})]$$

OR $E_{\eta}[T_j(\mathcal{X})] = T_j(\eta) \quad j=1, \dots, k$

$$\boxed{E_{\eta}[T(\mathcal{X})] = T(\eta)}$$

NOTE: CAN BE VIEW AS METHOD OF MOMENTS APPLIED TO NATURAL SUFFICIENT STATISTIC.

HOMEWORK: SHOW

$$i) \frac{\partial^2 d_0(\eta)}{\partial \eta_i \partial \eta_j} = -\text{COV}(T_i(\mathcal{X}), T_j(\mathcal{X}))$$

NOTE: ALTERNATIVE PROOF OF

$$ii) \mathbf{a}' H \mathbf{a} = -\text{VAR} \left[\sum_{j=1}^k a_j T_j(\mathcal{X}) \right] < 0$$

STRICT CONCAVITY OF log-likelihood,

COMMENT: (i) CAN BE USED TO SHOW $\Psi'(\eta) = \text{VAR}(\log \mathcal{X})$ when $\mathcal{X} \sim \text{GAMMA}(\alpha, 1)$

EXAMPLE: $X_1, \dots, X_n$ iid $\text{Beta}(\alpha, \beta)$

$$f(\underline{x}; \alpha, \beta) = B(\alpha, \beta)^{-n} \prod_{i=1}^n x_i^{\alpha-1} (1-x_i)^{\beta-1}$$

$$= \exp \left\{ (\alpha-1) \sum_{i=1}^n \log x_i + (\beta-1) \sum_{i=1}^n \log (1-x_i) - n \log B(\alpha, \beta) \right\}$$

$$T_1(\underline{x}) = \sum_{i=1}^n \log x_i, \quad T_2(\underline{x}) = \sum_{i=1}^n \log (1-x_i)$$

NOW $E[\log X] = \overline{\log X}$ AND $E[\log(1-X)] = \overline{\log(1-X)}$ ESTIMATING EQUATIONS

HOWEVER: $E[\log X] = \frac{1}{B(\alpha, \beta)} \int_0^1 (\log x) x^{\alpha-1} (1-x)^{\beta-1} dx$ COMPLICATED

NOTE: $\nabla \log B(\alpha, \beta) = (E[\log X], E[\log(1-X)])$

EXAMPLE: $X_1, \dots, X_n$ iid $N(\mu, \sigma^2)$

NAT. INV. SUFF. STAT: $T_1(\underline{x}) = \sum_{i=1}^n X_i, \quad T_2(\underline{x}) = \sum_{i=1}^n X_i^2$

ESTIMATING EQUATIONS: $\sum_{i=1}^n X_i = E[\sum_{i=1}^n X_i] = n\mu$

$$\sum_{i=1}^n X_i^2 = E[\sum_{i=1}^n X_i^2] = n(\mu^2 + \sigma^2)$$

$$\Rightarrow \hat{\mu} = \bar{X}, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \text{ IS UNIQUE MLE.}$$

OTHER PROPERTIES:

THEOREM 4.3. A MINIMAL SUFFICIENT STATISTIC FROM A FULL RANK EXPONENTIAL FAMILY IS COMPLETE.

PROOF: SEE THEOREM 1, p. 142, LEHMANN, E. (1986), TESTING STATISTICAL HYP., 2ND ED.

REVIEW: $X \sim F_{\theta}$, $\theta \in \mathcal{H}$. $g: \mathbb{R}^n \rightarrow \mathbb{R}$ IS COMPLETE IFF
(FROM 593) $E_{\theta}[g(X)] = 0 \Rightarrow g(X) = 0$ W.P.1 FOR ANY $\theta \in \mathcal{H}$

IMPORTANCE OF COMPLETENESS: LEHMANN-SCHNEFFE LEMMA

IF $T(X)$ IS COMPLETE AND SUFFICIENT, THEN $G(T(X))$ (real valued)
IS UNIQUE UMVUE OF $g(\theta) = E_{\theta}[G(T(X))]$: $h: \mathcal{H} \rightarrow \mathbb{R}$

CONSTRUCTION: RAO-BLACKWELL & THEOREM

$G(T(X)) = E_{\theta}[H(X) | T(X)]$ IS UNIQUE UMVUE FOR
 $h(\theta) = E_{\theta}[H(X)] = E_{\theta}[G(T(X))]$.

EXAMPLES: $X_1, \dots, X_n$ iid $N(\mu, \sigma^2)$

$T_1(X) = \sum X_i$, $T_2(X) = \sum X_i^2$ MINIMAL SUFFICIENT AND COMPLETE

$\Rightarrow \bar{X} = \frac{1}{n} \sum X_i$, $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ MINIMAL SUFFICIENT AND COMPLETE

$\Rightarrow \bar{X}$ UMVUE FOR μ and S^2 UMVUE FOR σ^2

REVIEW
(FROM 593) $\bullet$ ANY FUNCTION OF COMPLETE STATISTIC IS COMPLETE
 $\bullet$ ANY 1-1 FUNCTION OF MINIMAL SUFFICIENT IS MINIMAL SUFFICIENT

EXAMPLE: $X_1, \dots, X_n \text{ iid } N(\theta, \theta^2)$ (CURVED EXPONENTIAL FAMILY)

$(\bar{X}, S^2)$ IS MINIMAL SUFFICIENT BUT NOT COMPLETE

$$\text{SINCE } E[S^2] = \theta^2, E[\bar{X}^2] = \frac{\theta^2}{n} + \theta^2 = \left(\frac{n+1}{n}\right)\theta^2$$

$$\Rightarrow E[g(\bar{X}, S^2)] = 0 \text{ for } g(\eta, y) = \frac{n}{n+1}\eta^2 - y^2.$$

EXAMPLE: $X_1, \dots, X_n \text{ iid } N(\theta, \theta)$ $\theta > 0$ (FULL RANK EXPONENTIAL FAMILY)

$\Rightarrow \sum X_i^2$ IS COMPLETE SUFFICIENT

$\Rightarrow$ UNIQUE UMVUE OF θ EXISTS (SINCE UNBIAS ESTIMATES EXIST)

TO FIND $\hat{\theta}$, FIND h s.t. $E[h(\sum X_i^2)] = \theta$?

$\hat{\theta}$: FIND $h(\sum X_i^2) = E[\bar{X} | \sum X_i^2]$ CLOSED FORM?